Yeyuan (Yinzi) Xin
Professional Website Google Scholar Profile Github: yinzi-xin

Education

•PhD in Physics (2025)

California Institute of Technology (GPA: 4.0/4.0)

Thesis: Use of light coherence for exoplanet detection and characterization

Advisor: Dimitri Mawet

•MS in Aeronautics and Astronautics (2020)

Massachusetts Institute of Technology (GPA: 5.0/5.0)

Thesis: Coronagraphic data post-processing using projections on instrumental modes

Advisor: Kerri Cahoy

•BS in Physics (2018)

California Institute of Technology (GPA: 3.70/4.0)

Professional Research Experience

•01/10/2025-Present: Postdoctoral Researcher

Leiden University, Leiden, Netherlands

Advisor: Sebastiaan Haffert

-Investigating quantum-optimal coronagraph designs for the direct detection and spectroscopy of exoplanets. Developing astrophotonic hardware prototypes based on multi-plane light converters (also known as diffractive optical neural networks).

-Developing information-optimal approaches to coronagraphs, instrument control, instrument calibration, and statistical inference from complex optical astronomical data.

-Contributing to the coronagraphy and wavefront sensing and control work package for the Planetary Camera and Spectrograph (PCS) for the Extremely Large Telescope.

•01/10/2020-13/06/2025: PhD Student Researcher

California Institute of Technology, Pasadena, United States

Advisor: Dimitri Mawet

-Developed the photonic lantern nuller (PLN) instrument concept for the detection, imaging, and spectroscopy of close-in exoplanets. Characterized a PLN in the laboratory in monochromatic and broadband light. Demonstrated focal-plane wavefront control with the PLN. Tested and demonstrated PLN operations on-sky at the Subaru Telescope.

-Instrument development and daytime calibration team member of the Keck Planet Imager and Characterizer (KPIC) instrument at the Keck II telescope. Implemented a novel starlight suppression method on KPIC and validated algorithm on-sky. Investigated coherent differential imaging methods for improving detection sensitivity of high-resolution spectroscopy.

•01/02/2020-29/05/2020: Visiting Graduate Researcher

Space Telescope Science Institute, Baltimore, United States

Advisor: Laurent Pueyo

-Developed and characterized an instrument-aware data analysis method for improving exoplanet detection sensitivity from coronagraphic data.

-Investigated behavior of constraints in apodized vortex coronagraph optimization.

•01/07/2018-29/05/2020: Masters Student Researcher

Massachusetts Institute of Technology, Cambridge, United States

Advisor: Kerri Cahoy

-Aligned optical payload of Deformable Mirror (DeMi) demonstration cubesat. Wrote wavefront control software successfully used in orbit to close the adaptive optics loop.

-Simulated diffraction properties of laser guide star enabled segment cophasing for large, segmented space telescopes.

•15/06/2017-15/06/2018 Undergraduate Researcher

California Institute of Technology, Pasadena, United States

Advisor: Dimitri Mawet

-Implemented speckle nulling algorithm and Kalman filter for starlight suppression through a fiber injection unit. Helped integrate the High Contrast Spectroscopy Testbed.

Publications Overview

- First author of 5 refereed publications, 1 manuscript under review, 1 refereed review chapter, and 6 conference proceedings
- Second author of 1 student-led refereed publication
- Second or third author of 8 other refereed publications

h-index (Google Scholar): 19 Total citations (Google Scholar): 1117

See attached Publication List

Awards and Fellowships

•University of Arizona Alan Brass Prize Postdoctoral Fellowship (declined 2025).

•NASA Postdoctoral Program Fellowship (declined 2025).

•Judith G. Cohen Fellowship (2024), for work in diversity, equity, inclusion, and outreach in Caltech's Division of Physics, Math, and Astronomy. Funding awarded to design a survey to examine summer undergraduate student researcher recruitment methods at Caltech and identify ways to expand recruitment reach and increase diversity through the program.

•Honoree of Electro Optics Annual Photonics100 List (2024).

•National Science Foundation Graduate Research Fellowship (2018).

Scientific Organization

•Member of the science organizing committee for the Space-Based High Contrast Imaging in Europe III workshop (Edinburgh 2026).

•Founder and first organizer of the Leiden High Contrast Imaging Journal Club (November 2025-July 2026).

•Member of the Diversity, Equity, and Inclusion Committee in Caltech's Division of Physics, Math, and Astronomy (September 2024-May 2025).

Invited Talks (5 total):

•ASTRON (Netherlands Institute for Radio Astronomy) Colloquium (October 2026). *What is the best measurement of astronomical light? Quantum limits, spatial modes, and exoplanet imaging.*

•Peking University Astronomical and Biomedical Imaging Group Seminar (March 2026). *Optimal coronagraphy at the diffraction limit.*

•NASA Goddard General Meeting Presentation (virtual, November 2024). *Seeing within the inner working angle of coronagraphs with a Photonic Lantern Nuller.*

•University of California Santa Cruz Planetary Sciences Lunch Talk (October 2024). *How to be a light DJ: looking at exoplanets by remixing light*.

•University of Arizona High Contrast Imaging Seminar Series (April 2024). *Use of light coherence for exoplanet detection and characterization*.

Contributed Talks (10 total):

•*The Photonic Lantern Nuller: from concept to laboratory and on-sky demonstrations.* 2026. SPIE Astronomical Telescopes and Instrumentation, Copenhagen.

•*Photonic Lanterns for Astronomical Nulling Interferometry.* 2025. Workshop on Innovative Technologies for Space Optics, Noordwijk.

•*Seeing within the inner working angle of coronagraphs at infrared wavelengths with Photonic Lantern Nulling*. 2025. Towards the Habitable Worlds Observatory: Visionary Science and Transformational Technology, Washington, DC.

•*Wavefront sensing and control for a photonic lantern nuller for exoplanet characterization.* 2024. SPIE Astronomical Telescopes and Instrumentation, Yokohama.

•*Laboratory characterization of a mode-selective photonic lantern for exoplanet characterization*. 2023. SPIE Optics and Photonics, San Diego.

•*On-sky speckle nulling through a single-mode fiber with the Keck Planet Imager and Characterizer.* 2023. Adaptive Optics for the ELT, Avignon.

•*Prospects for reconstructing coherent starlight spectrum from KPIC speckle nulling data*. 2023. Coherence Differential Imaging Workshop, Paris.

•*Phase II of the Keck Planet Imager and Characterizer: On-sky Commissioning and Early Science Results*. 2022. Disks and Planets across ESO Facilities, Garching.

•*Coronagraphic Data Post-processing Using Projections on Instrumental Modes*. 2021. Network of Young Researchers in Instrumentation for Astronomy, virtual.

•*Diffraction analysis of Laser Guide Star enabled cophasing wavefront control for large segmented aperture space telescopes*. 2019. American Astronomical Society Meeting, Seattle.

Mentoring

•**Andrea Scardigno**, Leiden University Masters Research Project (2026-present)

Project: Spatial filtering using multi-plane light converters for the Large Interferometer for Exoplanets mission.

•**Joep Kraakman**, Leiden University Bachelors Research Programme (2026)

Project: Resolving exomoons using super-resolution imaging.

•**Suvinay Goyal**, Caltech Summer Undergraduate Research Fellow (2024, while undergraduate at University of Illinois Urbana-Champaign, now Caltech PhD student in Physics).
Project: Analysis of angular differential post-processing algorithms for exoplanet direct detection with the Photonic Lantern Nuller.

Teaching Experience

•2021: Teaching Assistant, Caltech, Classical Mechanics and Electromagnetism.

•2020-2021: Volunteer Tutor, Caltech, Rise Program.

•2018: Teaching Assistant, Caltech, Sophomore Physics Lab.

Outreach Activities

•Astronomy on Tap Leiden Organization Committee Member (2026-present).

•Astronomy on Tap Leiden Public Talk (October 2025). "The 3 Body Problem and Planet Habitability."

•Volunteer for Cambridge Science Week Drinking with the Stars Event (April 2019).

•Volunteer organizer for MIT High School Girls in Aerospace Day (October 2018).

Significant Software Contributions

•Lead developer and primary contributor to backPAC (back Propagating multi-plAne light Converter), code for optimizing multi-plane light converter phase screens based on autodifferentiable optical modeling with dLux.

•No. 2 contributor to pcs-cep (Planetary Camera and Spectrograph Coronagraph Evaluation Pipeline), for the evaluation of coronagraph designs for the PCS instrument.

Foreign Language Skills

Mandarin Chinese: elementary reading and writing, fluency in listening and speaking

Dutch: intermediate (A2 to B1) in reading, writing, listening, and speaking

French: lower-intermediate in reading and writing, elementary listening and speaking

Publication List

First-author Refereed Publications (2 in-prep, 1 under review, 5 published)

1. **Yinzi Xin**, Sebastiaan Haffert, Rico Landman, et al. Design and optimization of a mode-sorting coronagraph using multi-plane light converters. In-prep.

2. **Yinzi Xin**, Yoo Jung Kim, Jonathan Lin, et al. On-sky demonstration of the Photonic Lantern Nuller with the Subaru Coronagraphic Extreme Adaptive Optics Instrument. In-prep.

3. **Yinzi Xin**, Sebastiaan Haffert, Yoo Jung Kim, and Jonathan Lin. Quantum-optimal coronagraphy with spatial mode sorting for direct exoplanet observations. Under review at Astronomy and Astrophysics.

arXiv: https://doi.org/10.48550/arXiv.2607.02065

4. **Yinzi Xin**, Daniel Echeverri, Nemanja Jovanovic, et al. 2025. Implicit electric field conjugation with a Photonic Lantern Nuller. Journal of Astronomical Telescopes, Instruments, and Systems.

Publisher: https://doi.org/10.1117/1.JATIS.11.2.025004

arXiv: https://doi.org/10.48550/arXiv.2503.24292

5. **Yinzi Xin**, Daniel Echeverri, Nemanja Jovanovic, et al. 2024. Laboratory demonstration of a Photonic Lantern Nuller in monochromatic and broadband light. Journal of Astronomical Telescopes, Instruments, and Systems.

Publisher: https://doi.org/10.1117/1.JATIS.10.2.025001

arXiv: https://doi.org/10.48550/arXiv.2404.01426

6. **Yinzi Xin**, Laurent Pueyo, Romain Laugier, et al. 2024. Coronagraphic data post-processing using projections on instrumental modes. The Astrophysical Journal.

Publisher (open access): https://doi.org/10.3847/1538-4357/ad1879

7. **Yinzi Xin**, Jerry W. Xuan, Dimitri Mawet, et al. 2023. On-sky speckle nulling through a single-mode fiber with the Keck Planet Imager and Characterizer. Journal of Astronomical Telescopes, Instruments, and Systems.

Publisher: https://doi.org/10.1117/1.JATIS.9.3.035001

arXiv: https://doi.org/10.48550/arXiv.2307.11893

8. **Yinzi Xin**, Nemanja Jovanovic, Garreth Ruane, et al. 2022. Efficient detection and characterization of exoplanets within the diffraction limit: nulling with a mode-selective photonic lantern. The Astrophysical Journal.

Publisher (open access): https://doi.org/10.3847/1538-4357/ac9284

First-author Conference Proceedings (6 total)

•**Yinzi Xin**, Sebastiaan Haffert, and Rico Landman. 2026. Optimal mode-sorting coronagraphy: limits of single-moded measurements for the Habitable Worlds Observatory. SPIE Astronomical Telescopes and Instrumentation.

•**Yinzi Xin**, Nemanja Jovanovic, Dimitri Mawet, et al. 2026. The Photonic Lantern Nuller: from concept to laboratory and on-sky demonstrations. SPIE Astronomical Telescopes and Instrumentation.

•**Yinzi Xin**, Daniel Echeverri, Nemanja Jovanovic, et al. 2024. Wavefront sensing and control for a photonic lantern nuller for exoplanet characterization. SPIE Astronomical Telescopes and Instrumentation.

•**Yinzi Xin**, Daniel Echeverri, Nemanja Jovanovic, et al. 2023. Laboratory characterization of a mode-selective photonic lantern for exoplanet characterization. SPIE Optical Engineering and Applications.

•**Yinzi Xin**, Jerry W. Xuan, Dimitri Mawet, et al. 2023. On-sky speckle nulling through a single-mode fiber with the Keck Planet Imager and Characterizer. Adaptive Optics for Extremely Large Telescopes 7.

•**Yeyuan Xin**, Jorge Llop Sayson, Nikita Klimovich, et al. 2018. Demonstration of a speckle

nulling algorithm and Kalman filter estimator with a fiber injection unit for observing exoplanets with high-dispersion coronagraphy. SPIE Astronomical Telescopes and Instrumentation.

Refereed Scientific Reviews

•**Yinzi Xin**, Sergio Leon-Saval, and Eugene Serabyn. 2023. Nulling interferometry with optical fibers and photonic lanterns. Chapter 21 of the 2023 Astrophotonics Roadmap: pathways to realizing multi-functional integrated astrophotonic instruments. JPhys Photonics.
Publisher (open access): https://doi.org/10.1088/2515-7647/ace869

Student-led Refereed Publications

•Suvinay Goyal, **Yinzi Xin,** Nemanja Jovanovic, et al. Analysis of Angular-Differential Post-Processing Algorithms for Exoplanet Direct Detection with a Photonic Lantern Nuller. 2026. Journal of Astronomical Telescopes, Instruments, and Systems.
Publisher: https://doi.org/10.1117/1.JATIS.12.3.038001
arXiv: https://doi.org/10.48550/arXiv.2512.00442

Second- or Third-Author Publications (8 total)

•Luke Finnerty, **Yinzi Xin**, Jerry Xuan, et al. 2025. Water Dissociation and Rotational Broadening in the Atmosphere of KELT-20 b from High-resolution Spectroscopy. The Astronomical Journal.
•Luke Finnerty, **Yinzi Xin**, Jerry Xuan, et al. 2025. True mass and atmospheric composition of the nontransiting hot Jupiter HD 143105 b. The Astronomical Journal.
•Jonathan Lin, Michael Fitzgerald, **Yinzi Xin**, et al. 2025. Experimental and on-sky demonstration of spectrally dispersed wavefront sensing using a photonic lantern. Optics Letters.
•Luke Finnerty, Jerry Xuan, **Yinzi Xin**, et al. 2024. Atmospheric metallicity and C/O of HD 189733 b from high-resolution spectroscopy. The Astronomical Journal.
•Jonathan Lin, Michael Fitzgerald, **Yinzi Xin**, et al. 2023. Real-time experimental demonstrations of a photonic lantern wave-front sensor. The Astrophysical Journal Letters.
•Jonathan Lin, Michael Fitzgerald, **Yinzi Xin**, et al. 2023. Focal-plane wavefront sensing with photonic lanterns II: numerical characterization and optimization. Journal of the Optical Society of America B.
•Jonathan Lin, Michael Fitzgerald, **Yinzi Xin**, et al. 2022. Focal-plane wavefront sensing with photonic lanterns: theoretical framework. Journal of the Optical Society of America B.
•Benjamin Pope, Laurent Pueyo, **Yinzi Xin,** Peter Tuthill. 2021. Kernel phase and coronagraphy with automatic differentiation. The Astrophysical Journal.

Select Other Co-authored Publications

•Yoo Jung Kim, Michael Fitzgerald, Sebastien Vievard, Jonathan Lin, **Yinzi Xin**, et al. 2025. On-sky demonstration of subdiffraction-limited astronomical measurement using a photonic lantern. The Astrophysical Journal Letters.
•Yoo Jung Kim, Michael Fitzgerald, Jonathan Lin, Stephanie Sallum, **Yinzi Xin**, et al. 2024. Coherent imaging with photonic lanterns. The Astrophysical Journal.

•Jerry Xuan, Jason Wang, Luke Finnerty, et al. 2024. Validation of elemental and isotopic abundances in late-M spectral types with the benchmark HIP 55507 AB system. The Astrophysical Journal.
•Maïssa Salama, Charlotte Guthery, Vincent Chambouleyron, et al. 2024. Keck primary mirror closed-loop segment control using a vector-Zernike wavefront sensor. The Astrophysical Journal.
•Chih-Chun Hsu, Jason Wang, Jerry Xuan, et al. 2024. Rotation and abundances of the benchmark brown dwarf HD 33632 Ab from Keck/KPIC high-resolution spectroscopy. The Astrophysical Journal.
•Katelyn Horstman, Jean-Baptiste Ruffio, Konstantin Batygin, et al. 2024. RV measurements of directly imaged brown dwarf GQ Lup B to search for exosatellites. The Astronomical Journal.
•Sai Krishanth, Ewan Douglas, Ramya Anche, et al. 2024. Deepest limits on scattered light emission from the epsilon Eridani inner debris disk with HST/STIS. The Astronomical Journal.
•Daniel Echeverri, Jerry Xuan, John Monnier, et al. 2024. Vortex fiber nulling for exoplanet observations: first direct detection of M dwarf companions around HIP 21543, HIP 94666, and HIP 50319. The Astrophysical Journal Letters.
•Daniel Echeverri, Jerry Xuan, Nemanja Jovanovic, et al. 2023. Vortex fiber nulling for exoplanet observations: implementation and first light. Journal of Astronomical Telescopes, Instruments, and Systems.
•Maaike van Kooten, Sam Ragland, Rebecca Jensen-Clem, **Yinzi Xin**, et al. 2022. On-sky reconstruction of Keck primary mirror piston offsets using a Zernike wavefront sensor. The Astrophysical Journal.
•Rachel Morgan, Ewan Douglas, Gregory Allan, et al. 2021. Optical calibration and first light for the deformable mirror demonstration mission CubeSat (DeMi). Journal of Astronomical Telescopes, Instruments, and Systems.
•**See Google Scholar profile for full list of co-authored publications.**